AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT141
Introduction to Probability and Statistics

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21 / April /2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 90minutes**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions:4
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 30% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

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Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 1.5 marks, giving a total of 6 possible marks.

Q–1: You have an equal likely chance of choosing any number from 1 to 10. What is the probability that you can choose a number greater than 6?

a) 1/10 **c)** 1/2
b) 2/5 **d)** 3/5

Q–2: Events A and B are dependent P(A) = 50% and P(B|A) = 50% what is P(A ∩ B)?

a) 25% **c)** 50%
b) 100% **d)** 75%

Q–3: P(A) = 0.7, P(B) = 0.23, and P(A ∩ B) = 0.1, what is P(A U B)?

a) 0.83 **c)** 0.47
b) 0.13 **d)** 0.6

Q–4: In the array of scores 6, 12, 26, 32, 36, 48, 66, 94, 94 the score of 36 is:

a) The mean **c)** The mode
b) The median **d)** All of the above

Part 2: ESSAY QUESTIONS

Each question is worth 8 marks. You should answer all the three questions, giving a total of 24 possible marks.

Q–1: [2+2+1+3 Marks]

A sample of 10 young men joining airline hosting had their heights measured the heights were (in meter):

Heights (meter)	1.7	1.72	1.72	1.75	1.78	1.78	1.8	1.8	1.8	1.94

a) Calculate the sample mean for these data.
b) Calculate the sample median and mode.
c) Find the sample range.
d) Calculate the sample standard deviation.

Q-2: [2*4 Marks]

A company surveyed **100 employees** about their preferred work environment, categorized by department. The results are summarized below:

	Work From Home	Work From Office	Hybrid	Total
IT	30	10	10	50
HR	10	25	15	50
Total	40	35	25	100

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a) If an employee is selected at random, what is the probability that they are from the **IT department?**
b) If an employee is selected at random, what is the probability that they prefer **Hybrid** work?
c) If an employee is selected at random, what is the probability that they are from **HR AND prefer Work From Home?**
d) If an employee is selected at random, what is the probability that they prefer **Work From Office GIVEN that they are from HR?**

Q–3: [2+3+3 Marks]

Let X be a discrete random variable representing the number of cars sold per day. The probability distribution is given below:

X	0	1	2	3	4
P (X = x)	0.1	0.2	m	0.3	0.1

a) **Find the value of m.**
b) **Find P(X ≤ 2).**
c) **Find P(X > 1).**

End of questions

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